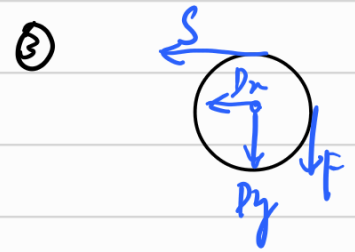
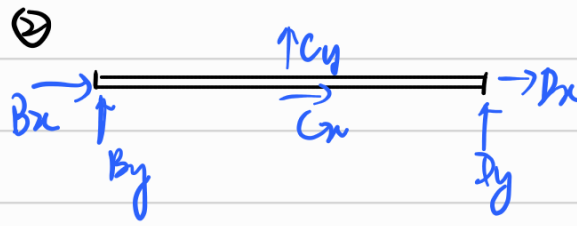
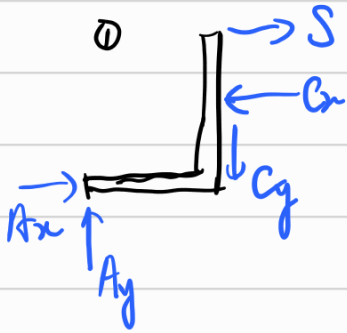
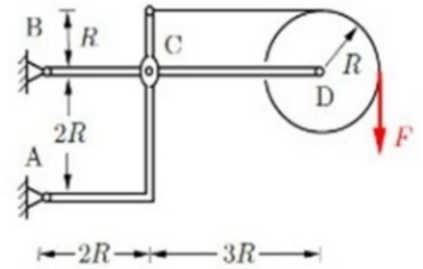


Homework 1.1 Determine the support reactions of depicted system, at A, B, C and D.



$$+\circlearrowleft M_A: 2RC_x - 2RC_y - 3RS = 0 \quad +\circlearrowleft M_D: -3RC_y - 5RB_y = 0$$

$$A_x + S = C_x$$

$$A_y = C_y$$

$$3C_y = 5B_y$$

$$\rightarrow \Sigma F_x: B_x + C_x + D_x = 0$$

$$B_x + C_x - F = 0$$

$$2C_x - 2C_y - 3F = 0$$

$$C_x - C_y = \frac{3}{2}F$$

$$C_x = 4F$$

$$A_x = 4F - F = 3F$$

$$+\uparrow \Sigma F_y: B_y + C_y - F = 0$$

$$B_y - \frac{5}{3}B_y = F$$

$$B_y = -\frac{3}{2}F$$

$$B_x = -3F$$

$$C_y = \frac{5}{3} \left(-\frac{3}{2} \right) F$$

$$C_y = -\frac{5}{2}F = A_y$$

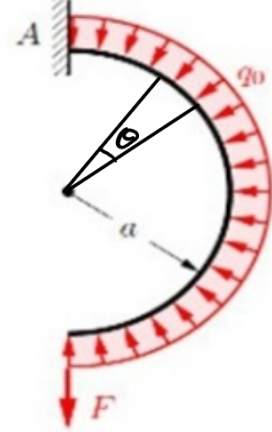
$$+\circlearrowleft M_D: RS - RF = 0 \Rightarrow S = F$$

$$+\uparrow \Sigma F_y: -D_y - F = 0 \Rightarrow D_y = -F$$

$$+\rightarrow \Sigma F_x: -D_x - S = 0 \Rightarrow D_x = -S$$

$$D_x = -F$$

Homework 1.2 A semicircular arc with the radius a is loaded by a radial line load q_0 and a vertical force F .



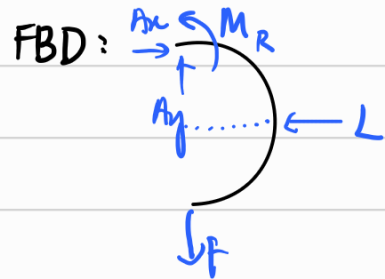
Determine the support reactions.

$$l = a d\theta \quad dq = q_0 dl$$

Should have a cos theta here

$$dq = q_0 a d\theta$$

$$\text{Total distributed load: } L = \int_{\theta=0}^{\theta=\pi} dq = \int_{\theta=0}^{\theta=\pi} q_0 a d\theta = q_0 a \theta \Big|_0^\pi = q_0 a \pi$$



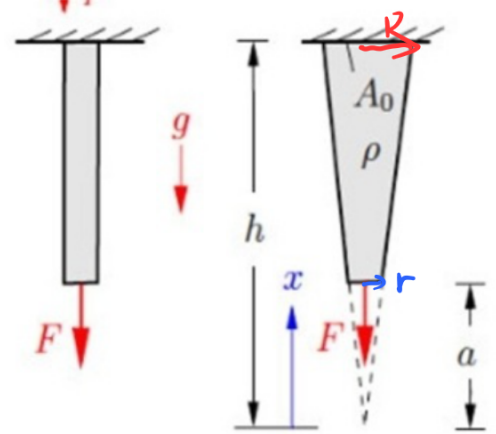
$$\begin{aligned} \uparrow + \sum F_y: A_y - F &= 0 \\ A_y &= F \end{aligned}$$

$$\begin{aligned} \rightarrow + \sum F_x: A_x - L &= 0 \\ A_x &= L = q_0 a \pi \end{aligned}$$

$$\sum \mathcal{M}_A = -La + M_R = 0$$

$$M_R = La = q_0 a^2 \pi$$

Homework 1.3 Determine the stresses distribution $\sigma(x)$ in the homogeneous bar due to its weight. The bar has constant thickness and a linear varying width. Furthermore, identify the location and value of the smallest stress.



$$\sigma(x) = \frac{N(x)}{A(x)} \quad A(x) = \frac{A_0}{h} x \quad A(x) = A_0 \frac{x}{h}$$

$$\rho = \frac{m}{V} \quad \rho = \frac{W}{Vg} \quad W(x) = \rho V(x) g = \rho g \int_a^x A_0 \frac{x}{h} dx$$

$$= \rho g \left(A_0 \frac{x^2}{2h} \right) \Big|_a^x$$

$$= \rho g A_0 \left(\frac{x^2 - a^2}{2h} \right)$$

$$N(x) = F + W(x) = F + \rho g A_0 \left(\frac{x^2 - a^2}{2h} \right)$$

$$\sigma(x) = \frac{N(x)}{A(x)} = \frac{F + \rho g A_0 \left(\frac{x^2 - a^2}{2h} \right)}{A_0 \frac{x}{h}} \times h$$

$$= \frac{Fh + \rho g \frac{A_0}{2} (x^2 - a^2)}{A_0 x}$$

To get minimum stress, we take the derivative of stress σ' to be 0.

$$\sigma'(x) = -\frac{Fh}{A_0} \frac{1}{x^2} + \frac{\rho g}{2} \left(1 + \frac{a^2}{x^2} \right) = 0 \Rightarrow z = \sqrt{\frac{2Fh}{\rho g A_0} - a^2}$$

$$\sigma_{\min} = \sigma(z) = \frac{Fh + \rho g \frac{A_0}{2} (z^2 - a^2)}{A_0 z}$$

$$= \frac{Fh}{A_0 z} + \frac{\rho g A_0}{2 A_0 z} (z^2 - a^2)$$

$$= \rho g \left(\frac{Fh}{\rho g A_0 z} + \frac{z^2 - a^2}{2z} \right)$$

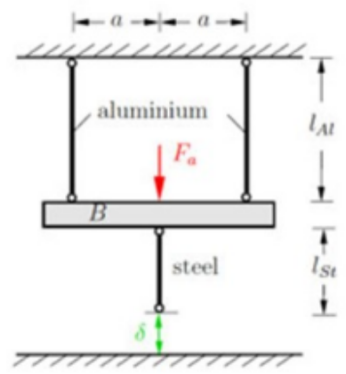
$$= \frac{\rho g}{z} \left(\frac{Fh}{\rho g A_0} + \frac{2Fh}{\rho g A_0} - a^2 \right)$$

$$= \frac{\rho g}{z} \left(\frac{Fh}{\rho g A_0} + \frac{Fh}{\rho g A_0} - a^2 \right)$$

$$= \frac{\rho g}{z} \left(\frac{2Fh}{\rho g A_0} - a^2 \right)$$

$$= \frac{\rho g}{z} (z^2) = \rho g z$$

Homework 1.4 In the depicted support construction for the *rigid* body B the lower support bar is too short by the length δ . In order to assemble the structure a force F_a is applied, such that the end of the bar just touches the ground. After assembly the force F_a is removed. The diameters of all bars d_i are identical.



a) Compute the required assembly force F_a .

b) Determine the displacement v_B of the body and the forces in the bars after assembly.

Given: $l_{Al} = 1\text{m}$, $d_{Al} = 2\text{mm}$, $E_{Al} = 70\text{GPa}$, $l_{St} = 1.5\text{m}$, $d_{St} = 2\text{mm}$, $E_{St} = 210\text{GPa}$, $\delta = 5\text{mm}$.

$$a) \Delta l = \frac{Fl}{EA} \quad \text{since } \delta = 5\text{mm} \Rightarrow \Delta l_{Al} + \Delta l_{St} = 5\text{mm}$$

F_a is equally distributed among the two aluminium bars

$$\frac{\left(\frac{F_a}{2}\right)(1 \times 10^3)}{70 \times 10^9 \times \pi (1 \times 10^{-3})^2} + \frac{F_a (1.5 \times 10^3)}{210 \times 10^9 \times \pi (1 \times 10^{-3})^2} = 5\text{mm}$$

Imagine the loading conditions. The steel bar has no loads here.

$$2.2736 \times 10^{-3} F_a + 2.2732 \times 10^{-3} F_a = 5$$

$$F_a = \frac{5}{4.5468 \times 10^{-3}} = 1099.5574\text{N} \approx 1.1\text{kN}$$

$$b) \Delta l_{Al} = \frac{1099.5574 \times 1 \times 10^3}{\pi (1 \times 10^{-3})^2 \times 70 \times 10^9} = 5\text{mm}$$

Now the F_a is withdrawn. Forces are generated by elongations and $2F_{Al} = F_{St}$ to reach the equilibrium.

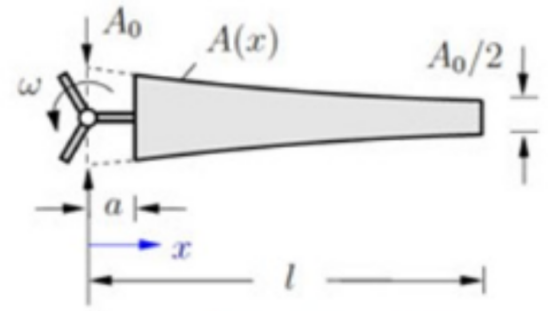
Hence, the displacement of body B ' v_B ' = 5mm

Homework 1.5 The cross section of a solid helicopter blade (density ρ , Young's modulus E) is described by the equation $A(x) = A_0 e^{-\alpha x/l}$. Determine the stress distribution $\sigma(x)$, if the blade is rotating with a constant angular velocity ω . Compute the elongation Δl under the assumption $a = 0$.

$$A(l) = \frac{A_0}{2} = A_0 e^{-\alpha} \quad \alpha = \ln 2$$

$$e^{-\alpha} = \frac{1}{2}$$

$$e^{\alpha} = 2$$

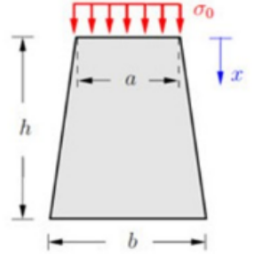


I'm sorry Prof. I tried but couldn't solve this Q.

Yeah the calculation is difficult. I will send you the answer and you can try it again.

Homework 1.6 A homogeneous frustum of a pyramid (Young's modulus E) with a square cross section is loaded on its top surface by a stress σ_0 .

Determine the displacement field $u(x)$ of a cross section at position x .



$$\sigma_0 = \frac{N}{A} \quad -\sigma_0 a^2 = N \text{ (compressive stress)} \quad \sigma = E\varepsilon \quad \varepsilon = \frac{\sigma}{E} \quad \frac{\Delta l}{l} = \frac{\sigma}{E} \quad \Delta l = \frac{\sigma l}{E} \quad \Delta l = \frac{N l}{EA}$$

$$\varepsilon = \frac{du}{dx} \quad \frac{N}{EA} = \frac{du}{dx}$$

$$A(x) = \left[(b-a)\frac{x}{h} + a \right]^2$$

$$-\sigma_0 a^2 = EA(x) \frac{du}{dx}$$

$$-\sigma_0 a^2 = E \left[a + (b-a)\frac{x}{h} \right]^2 \frac{du}{dx}$$

$$du = -\frac{\sigma_0 a^2}{E} \frac{dx}{\left[a + \left(\frac{b-a}{h} \right) x \right]^2}$$

$$\int_{u(0)}^{u(x)} du = -\frac{\sigma_0 a^2}{E} \int_0^x \frac{1}{\left[a + \left(\frac{b-a}{h} \right) x \right]^2} dx$$

$$\text{let } v = a + \left(\frac{b-a}{h} \right) x$$

$$\frac{dv}{dx} = \left(\frac{b-a}{h} \right)$$

$$dx = \frac{dv}{\frac{b-a}{h}}$$

$$u(x) - u(0) = -\frac{\sigma_0 a^2}{E} \int_a^{\left(\frac{b-a}{h} \right) x + a} \frac{1}{v^2} \left(\frac{b-a}{h} \right) dv$$

$$u(x) - u(0) = -\frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(-\frac{1}{v} \right) \Big|_a^{\left(\frac{b-a}{h} \right) x + a}$$

$$u(x) - u(0) = -\frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(\frac{1}{a} - \frac{1}{\left(\frac{b-a}{h} \right) x + a} \right)$$

$$\text{since } u(h) = 0 \Rightarrow u(h) - u(0) = -\frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$0 - u(0) = -\frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(\frac{b-a}{ab} \right)$$

$$u(0) = \frac{\sigma_0 a^2}{E} \left(\frac{h}{ab} \right) = \frac{\sigma_0 ah}{Eb}$$

$$u(x) = \frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(-\frac{1}{a} + \frac{1}{\left(\frac{b-a}{h} \right) x + a} + \frac{1}{a} - \frac{1}{b} \right)$$

$$= \frac{\sigma_0 a^2}{E} \left(\frac{h}{b-a} \right) \left(\frac{1}{\left(\frac{b-a}{h} \right) x + a} - \frac{1}{b} \right)$$